MAT 2040 Linear Algebra

# Assignment 8

Please note that

- Released date: 2024/07/11
- Due: 2024/07/19
- Late submission is NOT acceptable.
- Please submit your assignment as a PDF file titled "student ID + HW8".

**Question 1.** Which of the following sets of vectors form an orthonormal basis for $\mathbb{R}^2$ ?

(a) $\{(1,0)^T, (0,1)^T\}$

(b) $\left\{\left(\frac{3}{5}, \frac{4}{5}\right)^T, \left(\frac{5}{13}, \frac{12}{13}\right)^T\right\}$

(c) $\{(1,-1)^T, (1,1)^T\}$

(d) $\left\{\left(\frac{\sqrt{3}}{2}, \frac{1}{2}\right)^T, \left(-\frac{1}{2}, \frac{\sqrt{3}}{2}\right)^T\right\}$

**Question 2.** Let

$$\mathbf{u}_1 = \begin{pmatrix} \frac{1}{3\sqrt{2}} \\ \frac{1}{3\sqrt{2}} \\ -\frac{4}{3\sqrt{2}} \end{pmatrix}, \quad \mathbf{u}_2 = \begin{pmatrix} \frac{2}{3} \\ \frac{2}{3} \\ \frac{1}{3} \end{pmatrix}, \quad \mathbf{u}_3 = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \\ 0 \end{pmatrix}$$

(a) Show that $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ is an orthonormal basis for $\mathbb{R}^3$.

(b) Let $\mathbf{x} = (1,1,1)^T$. Write $\mathbf{x}$ as a linear combination of $\mathbf{u}_1$, $\mathbf{u}_2$, and $\mathbf{u}_3$ and compute $\|\mathrm{x}\|$.

**Question 3.** Let $S$ be the subspace of $\mathbb{R}^3$ spanned by the vectors $\mathbf{u}_2$ and $\mathbf{u}_3$ of Question 2. Let $\mathbf{x} = (1,2,2)^T$. Find the projection $\mathbf{p}$ of $\mathbf{x}$ onto $S$. Show that $(\mathbf{p} - \mathbf{x}) \perp \mathbf{u}_2$ and $(\mathbf{p} - \mathbf{x}) \perp \mathbf{u}_3$.

**Question 4.** Let $\mathbf{u}_1$ and $\mathbf{u}_2$ form an orthonormal basis for $\mathbb{R}^2$ and let $\mathbf{u}$ be a unit vector in $\mathbb{R}^2$. If $\mathbf{u}^T\mathbf{u}_1 = \frac{1}{2}$, determine the value of $\left|\mathbf{u}^T\mathbf{u}_2\right|$.

**Question 5.** Let $\{\mathbf{u}_1, \mathbf{u}_2, \mathbf{u}_3\}$ be an orthonormal basis for an inner product space $V$. If $\mathbf{x} = c_1\mathbf{u}_1 + c_2\mathbf{u}_2 + c_3\mathbf{u}_3$ is a vector with the properties $\|\mathbf{x}\| = 5, \langle \mathbf{u}_1, \mathbf{x}\rangle = 4$, and $\mathbf{x} \perp \mathbf{u}_2$, then what are the possible values of $c_1, c_2, c_3$ ?

**Question 6.** For each of the following, use the Gram-Schmidt process to find an orthonormal basis for Row$(A)$.

(a) $A = \begin{pmatrix} -1 & 3 \\ 1 & 5 \end{pmatrix}$

(b) $A = \begin{pmatrix} 2 & 5 \\ 1 & 10 \end{pmatrix}$

**Question 7.** Let $A = \begin{pmatrix} 2 & 1 \\ 1 & 1 \\ 2 & 1 \end{pmatrix}$ and $\mathbf{b} = \begin{pmatrix} 12 \\ 6 \\ 18 \end{pmatrix}$

(a) Use the Gram-Schmidt process to find an orthonormal basis for the column space of $A$.

(b) Factor $A$ into a product $QR$, where $Q$ has an orthonormal set of column vectors and $R$ is upper triangular.

(c) Solve the least squares problem $A\mathrm{x} = \mathrm{b}$

**Question 8.** Find the eigenvalues and the corresponding eigenspaces for each of the following matrices: (a) $\begin{pmatrix} 3 & 2 \\ 4 & 1 \end{pmatrix}$

(b) $\begin{pmatrix} 6 & -4 \\ 3 & -1 \end{pmatrix}$

(c) $\begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 0 & 0 & 0 \end{pmatrix}$

(d) $\begin{pmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 1 \end{pmatrix}$

(e) $\begin{pmatrix} 1 & 2 & 1 \\ 0 & 3 & 1 \\ 0 & 5 & -1 \end{pmatrix}$

**Question 9.** Let $A$ be an $n \times n$ matrix. Prove that $A$ is singular if and only if $\lambda = 0$ is an eigenvalue of $A$.

**Question 10.** Let $A$ be a nonsingular matrix and let $\lambda$ be an eigenvalue of $A$. Show that $1/\lambda$ is an eigenvalue of $A^{-1}$.

**Question 11.** An $n \times n$ matrix $A$ is said to be idempotent if $A^2 = A$. Show that if $\lambda$ is an eigenvalue of an idempotent matrix, then $\lambda$ must be either 0 or 1 .

**Question 12.** Let $A$ be an $n \times n$ matrix and let $B = A + I$. Is it possible for $A$ and $B$ to be similar? Explain.

**Question 13.** Let $Q$ be an orthogonal matrix.

(a) Show that if $\lambda$ is an eigenvalue of $Q$, then $|\lambda| = 1$.

(b) Show that $|\det(Q)| = 1$.

**Question 14.** Prove that, for any $\mathbf{u}$ and $\mathbf{v}$ in an inner product space $V$,

$$\|\mathbf{u} + \mathbf{v}\|^2 + \|\mathbf{u} - \mathbf{v}\|^2 = 2\|\mathbf{u}\|^2 + 2\|\mathbf{v}\|^2$$

Give a geometric interpretation of this result for the vector space $\mathbb{R}^2$.

**Question 15.** Consider the vector space $\mathbb{R}^n$ with inner product $\langle \mathbf{x}, \mathbf{y} \rangle = \mathbf{x}^T\mathbf{y}$. Show that for any $n \times n$ matrix $A$,

(a) $\langle A\mathbf{x}, \mathbf{y} \rangle = \langle \mathbf{x}, A^T\mathbf{y} \rangle$

(b) $\langle A^TA\mathbf{x}, \mathbf{x} \rangle = \|A\mathbf{x}\|^2$

**Question 16.** Let $\mathbf{u}$ be a unit vector in $\mathbb{R}^n$ and let $H = I - 2\mathbf{u}\mathbf{u}^T$. Show that $H$ is both orthogonal and symmetric and hence is its own inverse.

**Question 17.** Given $\mathbf{x}_1 = \frac{1}{2}(1, 1, 1, -1)^T$ and $\mathbf{x}_2 = \frac{1}{6}(1, 1, 3, 5)^T$, verify that these vectors form an orthonormal set in $\mathbb{R}^4$. Extend this set to an orthonormal basis for $\mathbb{R}^4$ by finding an orthonormal basis for the null space of

$$\begin{pmatrix} 1 & 1 & 1 & -1 \\ 1 & 1 & 3 & 5 \end{pmatrix}$$

[Hint: First find a basis for the null space and then use the Gram-Schmidt process.]

**Question 18.** Show that if two $n \times n$ matrices $A$ and $B$ have a common eigenvector $\mathbf{x}$ (but not necessarily a common eigenvalue), then $\mathbf{x}$ will also be an eigenvector of any matrix of the form $C = \alpha A + \beta B$.